

Relation Algebra

§1. Notation

Traditionally Adjunction $F \dashv G$ is defined as $F(x) \leq y$ iff $x \leq G(y)$, and you derive laws like this:

$$\begin{aligned} F(x) \leq F(x) &\implies x \leq G(F(x)) \implies F(x) \leq F(G(F(x))) \\ G(y) \leq G(y) &\implies F(G(y)) \leq y \implies G(F(G(y))) \leq G(y) \end{aligned}$$

By define $X : * \mapsto x$, $Y : * \mapsto y$ and use diagram order, application can be replaced by composition: $XF \leq Y$ iff $X \leq YG$

(1a)

¹ • write F as $/$, G as $\backslash$

² • render $a \leq b$ as $\boxed{\frac{a}{b}}$

³ • **rewrite** $X \leq XF$ as $1 \leq F$

$$\begin{aligned} \boxed{\frac{X/}{Y}} &= \boxed{\frac{X}{Y\backslash}} \\ \boxed{\frac{X/}{X/}} &\xRightarrow{\text{adj}} \boxed{\frac{X}{X\wedge}} \xRightarrow{\text{rewrite}} \boxed{\frac{*}{\wedge}}_{\text{unit}} \xRightarrow{\text{put } \backslash \text{ on left}} \boxed{\frac{\backslash}{\wedge\backslash}} \xRightarrow{\text{put } / \text{ on right}} \boxed{\frac{/}{\wedge}} \\ \boxed{\frac{Y\backslash}{Y\backslash}} &\xRightarrow{\text{adj}} \boxed{\frac{Y\vee}{Y}} \xRightarrow{\text{rewrite}} \boxed{\frac{\vee}{*}}_{\text{counit}} \xRightarrow{\text{put } / \text{ on left}} \boxed{\frac{\vee}{/}} \xRightarrow{\text{put } \backslash \text{ on right}} \boxed{\frac{\vee\backslash}{\backslash}} \end{aligned}$$

adj

now we have $\wedge = /$ and $\vee\backslash = \backslash$, and you just discovered string diagrams.

§1.1. Adjunctions

(1.1a)

	F ⊣ G	F's type	monad FG	1 ⊆ FG	GF ⊆ 1	F preserves u	G preserves n	FGF=F	GFG=G
1	$\triangleleft \dashv \triangleright$	$A \rightarrow A \otimes A$	$\triangleleft \triangleright = 1$	$1 \subseteq \triangleleft \triangleright$	$\triangleright \triangleleft \subseteq 1$	$(R \cup S) \triangleleft = R \triangleleft \cup S \triangleleft$	$(R \cap S) \triangleright = R \triangleright \cap S \triangleright$	$\triangleleft \triangleright \triangleleft = \triangleleft$	$\triangleright \triangleleft \triangleright = \triangleright$
2	$\multimap \dashv \multimap$	$A \rightarrow I$	$\multimap \multimap = T$	$1 \subseteq \multimap \multimap$	$\multimap \multimap \subseteq 1$	$(R \cup S) \multimap = R \multimap \cup S \multimap$	$(R \cap S) \multimap = R \multimap \cap S \multimap$	$\multimap \multimap \multimap = \multimap$	$\multimap \multimap \multimap = \multimap$
3	$\circ \dashv \circ$	$(A \rightarrow B) \rightarrow (B \rightarrow A)$	$R^{\circ \circ} = R$	$R^{\circ \circ} = R$	$R^{\circ \circ} = R$	$(R \cup S)^{\circ} = S^{\circ} \cup R^{\circ}$	$(R \cap S)^{\circ} = R^{\circ} \cap S^{\circ}$	$R^{\circ \circ} = R$	$R^{\circ \circ} = R$
4	$\circ \triangleleft \triangleright \multimap$	$(X \otimes A \rightarrow Y) \rightarrow (X \rightarrow Y \otimes A)$	1	$(\circ \triangleleft \otimes 1) (1 \otimes \triangleright \multimap) = 1$	$(1 \otimes \circ \triangleleft) (\triangleright \multimap \otimes 1) = 1$	=	=	=	=
5	$\triangleright \multimap \circ \triangleleft$	$(X \rightarrow Y \otimes A) \rightarrow (X \otimes A \rightarrow Y)$	1	$(1 \otimes \circ \triangleleft) (\triangleright \multimap \otimes 1) = 1$	$(\circ \triangleleft \otimes 1) (1 \otimes \triangleright \multimap) = 1$	=	=	=	=
6	$\Delta \dashv \cap$	$(A \rightarrow B) \rightarrow (A \rightarrow B)^2$	$R \cap R = R$	$R \cap R = R$	$R \cap S \subseteq R$	$\Delta(R \cup S) = \Delta R \cup \Delta S$	$(R \cap T) \cap (S \cap U) = (R \cap S) \cap (T \cap U)$	$R \cap R = R$	$R \cap R = R$
7	$\cup \dashv \Delta$	$(A \rightarrow B)^2 \rightarrow (A \rightarrow B)$	$(R, S) \mapsto (R \cup S, R \cup S)$	$R \subseteq R \cup S$	$R \cup R = R$	$(R \cup T) \cup (S \cup U) = (R \cup S) \cup (T \cup U)$	$\Delta(R \cap S) = \Delta R \cap \Delta S$	$R \cup R = R$	$R \cup R = R$
8	$\emptyset \dashv !$	$\{*\} \rightarrow (A \rightarrow B)$			$\emptyset \subseteq R$				
9	$\cdot S \dashv / S$	$(A \rightarrow B) \rightarrow (A \rightarrow C)$	S/S	$R \subseteq (RS)/S$	$(T/S)S \subseteq T$	$(S \cup T)R = (SR) \cup (TR)$	$(R \cap R') / S = (R / S) \cap (R' / S)$	$((RS)/S)S = RS$	$((T/S)S)/S = T/S$
10	$S \cdot \dashv S \setminus$	$(B \rightarrow C) \rightarrow (A \rightarrow C)$	$S \setminus S$	$R \subseteq S \setminus (SR)$	$S(S \setminus T) \subseteq T$	$R(S \cup T) = (RS) \cup (RT)$	$S \setminus (R \cap R') = (S \setminus R) \cap (S \setminus R')$	$S(S \setminus (SR)) = SR$	$S \setminus (S(S \setminus T)) = S \setminus T$
11	$R \cap \dashv R \Rightarrow$	$(A \rightarrow B) \rightarrow (A \rightarrow B)$	$X \mapsto R \Rightarrow (X \cap R)$	$X \subseteq R \Rightarrow (X \cap R)$	$R \cap (R \Rightarrow Y) \subseteq Y$	$R \cap (X \cup Y) = (R \cap X) \cup (R \cap Y)$	$R \Rightarrow S \cap T = (R \Rightarrow S) \cap (R \Rightarrow T)$	$R \cap (R \Rightarrow (X \cap R)) = X \cap R$	$R \Rightarrow (R \cap (R \Rightarrow Y)) = R \Rightarrow Y$
12	$\mathcal{D} \dashv \cdot T$	$(A \rightarrow B) \rightarrow \text{Cor } A$	$R \mapsto (\mathcal{D}R)T$	$R \subseteq (\mathcal{D}R)T$	$\mathcal{D}(AT) \subseteq A$	$\mathcal{D}(R \cup S) = \mathcal{D}R \cup \mathcal{D}S$	$(A \cap B)T = AT \cap BT$	$\mathcal{D}((\mathcal{D}R)T) = \mathcal{D}R$	$(\mathcal{D}(AT))T = AT$
13	$\mathcal{R} \dashv T \cdot$	$(A \rightarrow B) \rightarrow \text{Cor } B$	$R \mapsto T(\mathcal{R}R)$	$R \subseteq T(\mathcal{R}R)$	$\mathcal{R}(TA) \subseteq A$	$\mathcal{R}(R \cup S) = \mathcal{R}R \cup \mathcal{R}S$	$T(A \cap B) = TA \cap TB$	$\mathcal{R}(T(\mathcal{R}R)) = \mathcal{R}R$	$T(\mathcal{R}(TA)) = TA$
14	$\cdot f \dashv \cdot f^{\circ}$	$(A \rightarrow B) \rightarrow (A \rightarrow C)$	ff°	$\text{Map}(f) \Rightarrow 1 \subseteq f f^{\circ}$	$f^{\circ} f \subseteq 1$	$(S \cup T)R = (SR) \cup (TR)$	$\text{Simple}(g) \Rightarrow (X \cap Y)g^{\circ} = (Xg^{\circ}) \cap (Yg^{\circ})$	$ff^{\circ}f = f$	$f^{\circ}ff^{\circ} = f^{\circ}$
15	$f^{\circ} \cdot \dashv f \cdot$	$(A \rightarrow C) \rightarrow (B \rightarrow C)$	ff°	$\text{Map}(f) \Rightarrow 1 \subseteq f f^{\circ}$	$f^{\circ} f \subseteq 1$	$R(S \cup T) = (RS) \cup (RT)$	$f(X \cap Y) = fX \cap fY$	$ff^{\circ}f = f$	$f^{\circ}ff^{\circ} = f^{\circ}$
16	$i \dashv E$	$\text{Map} \multimap \text{Rel}$	E	$\frac{1}{\exists} : A \rightarrow PA$	$\exists : PB \rightarrow B$	—	—	$\frac{\exists}{\exists} = 1$	$\frac{R}{\exists} \exists = R$
17	$\cdot \exists \dashv \frac{\exists}{\exists}$	$\text{Map}(A, PB) \rightarrow (A \rightarrow B)$	1	$\frac{f \exists}{\exists} = f$	$\frac{R \exists}{\exists} = R$	—	—	$\frac{f \exists}{\exists} \exists = f \exists$	$\frac{R \exists}{\exists} \exists = R \exists$
18	$\frac{\exists}{\exists} \dashv \cdot \exists$	$(A \rightarrow B) \rightarrow \text{Map}(A, PB)$	1	$\frac{R \exists}{\exists} \exists = R$	$\frac{f \exists}{\exists} = f$	—	—	$\frac{\exists}{\exists} \exists = \frac{R}{\exists}$	$\frac{f \exists}{\exists} \exists = f \exists$

§1.2. Composing adjunctions

(1.2a)

$$\begin{array}{c}
\begin{array}{ccccc}
\boxed{\begin{array}{c} X \\ f(R/T) \end{array}} & \xLeftrightarrow{f^{\circ} \cdot \dashv f \cdot} & \boxed{\begin{array}{c} f^{\circ}X \\ R/T \end{array}} & \xLeftrightarrow{\cdot T \dashv / T} & \boxed{\begin{array}{c} (f^{\circ}X)T \\ R \end{array}} \\
\text{f a map} & & & & \\
\begin{array}{ccccc}
& \xLeftrightarrow{\text{associativity}} & \boxed{\begin{array}{c} f^{\circ}(XT) \\ R \end{array}} & \xLeftrightarrow{f^{\circ} \cdot \dashv f \cdot} & \boxed{\begin{array}{c} XT \\ fR \end{array}} & \xLeftrightarrow{\cdot T \dashv / T} & \boxed{\begin{array}{c} X \\ (fR)/T \end{array}} \\
& & & & & & \\
& & \xRightarrow{\text{indirect equality}} & \boxed{\begin{array}{c} f(R/T) \\ (fR)/T \end{array}} & & &
\end{array}
\end{array}$$

$$\text{Map}(f) \Rightarrow f(R/S) = (fR)/S$$

(1.2b)

	$\cdot T$	$T \cdot$	$\cdot g$	$g^\circ \cdot$	Tn	$^\circ$	$\circ \triangleleft$	Δ
1	$\cdot S$	$R / (S_1 \ S_2) = R / S_2 / S_1$	$S \setminus (R / W) = (S \setminus R) / W$	$R/(Sg) = (Rg^\circ)/S$	$\text{Map}(f) \Rightarrow f$ $(R / S) = (f R) / S$	$(Y/S)^\circ = S^\circ \setminus (Y^\circ)$	$(RS)^\wedge = R^\wedge (S \otimes 1)$	$(R \cap R') / S = (R / S) \cap (R' / S)$
2	$S \cdot$	$S \setminus (R / W) = (S \setminus R) / W$	$(S_1 \ S_2) \setminus R = S_2 \setminus (S_1 \setminus R)$	$S \setminus (Rg^\circ) = (S \setminus R)g^\circ$	$(g^\circ S) \setminus R = S \setminus (gR)$	$(S \setminus Y)^\circ = Y^\circ / S^\circ$	$((S \otimes 1)R)^\wedge = SR^\wedge$	$S \setminus (R \cap R') = (S \setminus R) \cap (S \setminus R')$
3	$\cdot f$	$\text{Map}(f) \Rightarrow R / (f S) = (R / S) f^\circ$	$T \setminus (Rf^\circ) = (T \setminus R) f^\circ$	$(R S)^\circ = S^\circ R^\circ$		$(R S)^\circ = S^\circ R^\circ$		$\text{Simple}(g) \Rightarrow (X \cap Y) g^\circ = (X g^\circ) \cap (Y g^\circ)$
4	$f^\circ \cdot$	$\text{Map}(f) \Rightarrow f (R / S) = (f R) / S$	$(Tf^\circ) \setminus R = f(T \setminus R)$		$(R S)^\circ = S^\circ R^\circ$	$(R S)^\circ = S^\circ R^\circ$		$f(T_1 \cap T_2) = fT_1 \cap fT_2$
5	Rn	—	—	—	$R \Rightarrow S \Rightarrow T = R \cap S \Rightarrow T$	$(R \Rightarrow Y)^\circ = R^\circ \Rightarrow (Y^\circ)$	—	$R \Rightarrow S \cap T = (R \Rightarrow S) \cap (R \Rightarrow T)$
6	$^\circ$	$(Y/T)^\circ = T^\circ \setminus (Y^\circ)$	$(T \setminus Y)^\circ = Y^\circ / T^\circ$	$(R S)^\circ = S^\circ R^\circ$	$(T \Rightarrow Y)^\circ = T^\circ \Rightarrow (Y^\circ)$	$R^\circ \circ = R$	$(R^\wedge)^\circ = (R^\circ \otimes 1)$ $(1 \otimes \triangleright \circ)$ both bends: the $^\circ$ section's display	$(T_1 \cap T_2)^\circ = T_1^\circ \cap T_2^\circ$
7	$\circ \triangleleft$	$(RT)^\wedge = R^\wedge (T \otimes 1)$	$((T \otimes 1)R)^\wedge = TR^\wedge$	—	—	$(R^\wedge)^\circ = (R^\circ \otimes 1)$ $(1 \otimes \triangleright \circ)$ both bends: the $^\circ$ section's display	—	—
8	u	$(S \cup T) R = (S R) \cup (T R)$	$R (S \cup T) = (R S) \cup (R T)$	$(S \cup T) R = (S R) \cup (T R)$	$R (S \cup T) = (R S) \cup (R T)$	$R \cap (S \cup T) = R \cap S \cup R \cap T$	$(R \cup S)^\circ = S^\circ \cup R^\circ$	
9	$\emptyset$	$\emptyset R = \emptyset$	$R \emptyset = \emptyset$	$\emptyset R = \emptyset$	$R \emptyset = \emptyset$	$R \cap \emptyset = \emptyset$	$\emptyset^\circ = \emptyset$	

§1.3. Adjoint triples

Beyond $u \dashv \Delta \dashv n$, which §Section 1.1 already carries as two rows, the table holds one adjoint triple with content: $\exists_f \dashv f^* \dashv \forall_f$ along a map f , once on each side of the composite. $f : A \rightarrow B$ throughout this subsection.

(1.3a)

$$\begin{aligned} \cdot f \dashv \cdot f^\circ \dashv /f^\circ \\ f^\circ \cdot \dashv f \cdot \dashv f \setminus \end{aligned}$$

The first two links of each chain are the map shunting rules, and the third is §Section 1.1's division row read at a chosen divisor: $\cdot S \dashv /S$ at $S := f^\circ$, and $S \cdot \dashv S \setminus$ at $S := f$.

The third link is not the first over again. For a map f , $/f$ collapses to $\cdot f^\circ$; being a map is exactly what that collapse spends $\text{Map}(f) \Rightarrow R / (f S) = (R / S) f^\circ$ — f° is not a map, so $/f^\circ$ does not collapse in turn, and the three operators stay distinct. On $\text{Rel}(\mathbb{I}, A) = PA$ they are the image triple:

(1.3b)

	operator	acts by	name
1	$\cdot f$	$S \mapsto f[S]$	direct image, $\exists_f$
2	$\cdot f^\circ$	$T \mapsto f^{-1}[T]$	inverse image, f^*
3	$/f^\circ$	$S \mapsto \{b : f^{-1}(b) \subseteq S\}$	$\forall_f$

§Section 1.1's $\mathcal{D} \dashv \cdot T$ is this same chain along the projection $A \otimes B \rightarrow A$, read through $\text{Rel}(A, B) = P(A \otimes B)$: $\mathcal{D}R = \{(a, a) : \exists b. aRb\}$, AT is the relation that ignores b altogether, and the third link is $R \mapsto 1 \cap R / T = \{(a, a) : \forall b. aRb\}$.

(1.3c)

$$\mathcal{D} \dashv \cdot T \dashv 1 \cap \cdot / T$$

Three is where it stops, and one map breaks both ends. Take $A=\{a_1, a_2\}$ and $B=\{b\}$: $f[\{a_1\}] \cap f[\{a_2\}] = \{b\}$ while $f[\{a_1\} \cap \{a_2\}] = \emptyset$, so $\cdot f$ does not preserve meets and has no left adjoint, and the same two subsets give $\bigvee f(\{a_1\} \cup \{a_2\}) = \{b\}$ against $\bigvee f\{a_1\} \cup \bigvee f\{a_2\} = \emptyset$, so $/f^\circ$ does not preserve joins and has no right adjoint. A monic f restores the binary case and no more — τf still reaches only $\text{im } f$, and $\bigvee f\emptyset = B \setminus \text{im } f$ is still not $\emptyset$. The chain extends only when f° is a map as well, and then $/f^\circ = \cdot f^\circ = \cdot f$ and it repeats forever. For an S with neither S nor S° a map there is no triple at all: $\cdot S \dashv S$ is two links and stops.

The rows that chain forever say nothing by it: $^\circ$ is an order-isomorphism of hom-posets, so it is adjoint to itself on both sides and $^\circ \dashv^\circ \dashv^\circ$ has no end, and §Section 1.1's other row of that kind, $\circ \triangleleft \dashv \triangleright \circ$, goes the same way.